DEON TAKYI-AMPONSEM YEBOAH

Portfolio: Github: https://github.com/Calibre-sys

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PROFESSIONAL SUMMARY

Self-taught software developer and published machine-learning researcher with a Petroleum Engineering background. I build AI-enabled products end to end, from unsupervised machine learning systems, published in a peer-reviewed journal, to full-stack ML web applications, to workflow automations. I specialize in turning slow, manual, domain-heavy processes into intelligent, scalable software.

TECHNICAL SKILLS

- **Languages:** Python (Fast API), JavaScript (Node.js, React.js, Google Apps Script), HTML/CSS, VBA
- **Machine Learning & Data Science:** supervised & unsupervised learning, deep learning, time-series forecasting, MLOps
- **Automation & Tooling:** Google Apps Script, Googleapis, Zapier, AI agents Development, financial modelling with VBA/Excel
- **Applied LLMS-**, MCPs, Context Windows, RAG.
- **Domain Software:** AutoCAD, Prosper, GAP, Eclipse 100 & 300 (petroleum engineering simulation)

EXPERIENCE

Business Development / Technical Associate (National Service) | SOH Energy, Ghana *Nov 2025 – Present*

- Built SOH Order Management system, a software that streamlines BDC operations by automating document generation, and email dispatch, tracking of orders, cutting processing time by over 90%.
- Built an AI-agent recruitment system that screens and ranks CVs automatically, replacing manual shortlisting and generating candidate emails end to end.
- Replaced a 900-contact manual phone-call campaign with a fully automated Google Apps Script email workflow — custom emails with one-click responses, automatic sheet logging, 3-day follow-up reminders, and thank-you confirmations, converting 455 of 880 contacts (52%) with zero manual calls.
- Co-developed a financial feasibility model for an urban fuel station, incorporating demand forecasting, cost analysis, and profitability analysis.

Reservoir Characterization Lead (ML Research) | i-DPRS Research Lab, KNUST *Sep 2024 – Present*

- Built RescharAI, an industry software that applies unsupervised ML for HFU identification and supervised ML with residual correctors for net-pay delineation.
- Co-developed a novel automated system for Hydraulic Flow Unit (HFU) identification from core and well data using autoencoders and unsupervised machine learning; published in the Journal of Applied Geophysics.
- Collaborated with Ghana National Petroleum Corporation (GNPC) to apply unsupervised machine learning to enhance reservoir characterization on real field data.
- Engineered a causal analysis tool for production engineers to determine the effects of water injectors on producers, improving

Operations Intern | Booster Station (Tema) & Tema Tank Farm, Chase Logistics *Sep 2023 – Jan 2024*

- Supported product tracking to terminals, flow-meter calibration with proval tanks, and periodic checks on pumps, valves, pipelines, and storage tanks — hands-on exposure to the operational data problems I now automate.

SELECTED PROJECTS

SOH Order Tracker - *Automated document software integrating Google Sheets and automated emails, reducing the processing time for orders and invoices by over 90%.*

RescharAI — *ML software for reservoir characterization*

- Designed and built software combining unsupervised learning (HFU identification) and supervised learning with residual correctors (net-pay delineation), removing the need for manual petrophysical workflows.

Fortel — *no-code machine-learning web application (www.fortel.onrender.com)*

- Co-developed a full-stack ML web app that lets users train models on securely uploaded private data without writing code; designed the entire front end (HTML, CSS, JavaScript) and am integrating React.js/Node.js for ongoing improvements.

Manpower Database Automation — *Google Apps Script, googleapis*

- End-to-end email automation with one-click accept buttons, automated sheet logging, timed follow-ups, and confirmation emails, turning a months-long calling task into a two-day build that ran itself.

PUBLICATION

- *Automated Hydraulic Flow Unit Determination Using FZI-Z-Score Probability, K-Means Clustering with Autoencoders, and Machine Learning Classification: A Case Study of the Subei Basin, China.* Journal of Applied Geophysics, 2025. https://doi.org/10.1016/j.jappgeo.2025.105778
- Analyzing the Effects of Water Injectors on Oil Production Using Causality Models. https://doi.org/10.2118/233427-MS

EDUCATION

BSc Petroleum Engineering | Kwame Nkrumah University of Science & Technology (KNUST) *2021 – 2025*

- KNUST College of Engineering Excellence Award, 2021/2022 academic year.

W.A.S.S.C.E (General Science) | Mfantsipim School, Cape Coast *2018 – 2021*

- 2nd Best Student, 2020/2021 academic year.

CERTIFICATIONS

- Introduction to Deep Learning
- Intermediate Deep Learning
- Time Series Forecasting — DataCamp
- Frontend Development- Codeacademy
- Build Chat Applications with LangGraph— 365 Data Science (ID: CC-EE76A0036)
- Machine Learning Algorithms A–Z — 365 Data Science (ID: CC-A58D17499)

Events & Positions

- Delegate at Africa Technology Conference 2026- Abidjan, Côte d'Ivoire.
- Building Bytes Pitch Deck Program, Kumasi
- Treasurer (Executive), Society of Petroleum Engineers — KNUST Chapter, 2024/2025; active SPE member since 2021.
- SPE Energy4me volunteer

REFEREES

Dr. Stephen Adjei, PhD — Principal Investigator, AI-DPRS; Lecturer, Dept. of Petroleum Engineering, KNUST
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Mr. Felix Andoh — Operator, Tema Tank Farm (TTF)
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